

ScholarShield: Confidential Scholarship & Eligibility Protocol

Midnight Network Rise In • Level 4 Comprehensive Idea Description & Technical Proposal

1. Executive Summary

ScholarShield is a decentralized, zero-knowledge scholarship eligibility verification and confidential academic credential protocol built on Midnight Network. It eliminates the need for students to disclose sensitive tax returns, household bank balances, or complete GPA transcripts to scholarship committees, while providing donors and universities with mathematical certainty of eligibility without data fraud.

2. Problem Landscape & Need for Zero-Knowledge

In traditional higher education philanthropy, millions of dollars in grants are awarded annually, but the verification process suffers from severe systemic privacy vulnerabilities:

- **Socioeconomic Vulnerability Exposure:** Students must disclose detailed tax filings and bank records to broad evaluation panels, leading to bias and privacy leaks.
- **Pervasive Application Fraud:** Forged PDF transcripts and duplicate submissions across multiple universities siphons funds away from deserving scholars.
- **Public Blockchain Data Leaks:** Existing Web3 credential protocols publish credentials on public chains, publicly exposing personal scores to block explorers.

3. Technical Architecture & Midnight Implementation

ScholarShield utilizes Midnight's **Compact 0.19** language to implement a multi-dimensional zero-knowledge verification circuit:

- **Need-Based Circuit Constraint:** Proves `privateIncomeUSD <= maxIncomeThresholdUSD` without disclosing the student's exact income.
- **Merit-Based Circuit Constraint:** Proves `privateAcademicPercentageBps >= minAcademicPercentageBps` without revealing grades.
- **University KYC Whitelist:** Cryptographically verifies student enrollment via Merkle tree membership without revealing student identity.
- **Anti-Double-Claim Nullifiers:** Enforces deterministic nullifiers `Hash(studentSecret, scholarshipId)` preventing duplicate claims.
- **Midnight Rational Privacy:** Supports selective disclosure viewing keys (`grantAuditorAccess`) for IRS / 501(c)(3) regulatory audits.

4. Zero-Knowledge Privacy Invariant Matrix

Data Field	Student Local	Midnight Ledger	Verifier Visibility	Privacy Status
Household Income (\$)	Visible (\$32,000)	Hidden (0 bytes)	Hidden (●●●●●●)	Private Witness
Academic Score (%)	Visible (92.5%)	Hidden (0 bytes)	Hidden (●●●●●●)	Private Witness
Student Secret Key	Visible (0x7a...)	Hidden (0 bytes)	Hidden (●●●●●●)	Private Seed
Program Criteria	Visible	Public (\$50k / 85%)	Visible	Public Rule

Application Nullifier	Visible	Public (0x8f...)	Visible (Auditable)	Public Anchor
Eligibility Decision	Visible	Public (ELIGIBLE)	Visible (ELIGIBLE)	Public Result

5. Deliverables & Alignment with Midnight RFPs

ScholarShield aligns directly with the **Identity, Compliance & Verifiable Credentials** and **Confidential Credentials** tracks. The repository contains a fully working Next.js 15 frontend, official Midnight Lace Wallet connector, 9 passing unit tests, and production CI/CD workflows.

- **GitHub Repository:** <https://github.com/shivang-tech-3/ScholarShield>
- **Live Web Application:** <https://scholarshieldmoonlight.netlify.app/>
- **Preprod Contract Address:** 0x71a4f89d02b84719283746501928374650192837465019283746501928374650